

HW01

ENAE 641: Linear System Dynamics

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Problem 1

a) Find the coordinates of the vector $\mathbf{x}^T = [-3, 2, 4]$ in the basis given by

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix} \quad \mathbf{v}_3 = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$$

b) Determine the rank and nullity of $\mathbf{A}$, and a basis for its range and nullspace when

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \\ 6 & 3 & 0 \end{bmatrix}$$

a) Coordinates in new basis

We would like scalars c_1, c_2, c_3 such that

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3$$

Construct a matrix with the basis vectors as columns:

$$\mathbf{V} = [\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] = \begin{bmatrix} 2 & -2 & 3 \\ -1 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix}$$

Set up change of basis equation:

$$[\mathbf{v}_1 \ \mathbf{v}_2 \ \mathbf{v}_3] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \mathbf{x}$$

Substituting known values:

$$\begin{bmatrix} 2 & -2 & 3 \\ -1 & 2 & 1 \\ 4 & 1 & 3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} -3 \\ 2 \\ 4 \end{bmatrix}$$

Row reducing:

$$\left[\begin{array}{ccc|c} 2 & -2 & 3 & -3 \\ -1 & 2 & 1 & 2 \\ 4 & 1 & 3 & 4 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{29}{31} \\ 0 & 1 & 0 & \frac{53}{31} \\ 0 & 0 & 1 & -\frac{15}{31} \end{array} \right]$$

Thus, relative to the ordered basis $\mathcal{B} = [\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3]$,

$$[\mathbf{x}]_{\mathcal{B}} = \frac{1}{31} \begin{bmatrix} 29 \\ 53 \\ -15 \end{bmatrix} \quad \blacksquare$$

b) Rank, range, nullity, and nullspace

Row reducing $\mathbf{A}$:

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 4 & 2 & 0 \\ 6 & 3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

There is one pivot and two free variables, so:

$$\text{rank}(\mathbf{A}) = 1 \quad \blacksquare$$

$$\text{nullity}(\mathbf{A}) = 2 \quad \blacksquare$$

Regarding the original $\mathbf{A}$ matrix, first column is a pivot column, the second column is half of the first (meaning it can be expressed as a linear combination of the first column), and the third is zero. Thus, the first column forms a basis for its range:

$$\mathbf{B}_{\text{range}(\mathbf{A})} = \left\{ \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \right\} \quad \blacksquare$$

To find the nullspace, we solve:

$$\mathbf{A}\mathbf{z} = \mathbf{0}$$

Substituting known values:

$$2z_1 + z_2 = 0$$

Choosing $z_2 = 2s$ and $z_3 = t$:

$$\mathbf{z} = \begin{bmatrix} -s \\ 2s \\ t \end{bmatrix} = s \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Thus, a basis for the nullspace is:

$$\mathbf{B}_{\text{nul}(\mathbf{A})} = \left\{ \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \quad \blacksquare$$

Problem 2

Using the simple pendulum example from the lectures, suppose $l = 2, m = 1$ (metric units), and $b'(0) = 8$.

- Determine the trim input τ_m^* needed to maintain an equilibrium angle $\theta^* = \frac{\pi}{3}$. Determine the corresponding input-output differential equation that models deviations of the system from this equilibrium.
- With the input held constant, $\tau_{m(t)} = \tau_m^*$ from above, suppose the pendulum is moved to an angle slightly offset from $\frac{\pi}{3}$ and released. Make a sketch of the resulting motion of pendulum $\theta(t)$. Identify oscillation frequency, damping ratio, and settling time of this response as appropriate.

a) Trim input and deviation equation

Using the supplied lecture model, $z_1 = \theta$, $z_2 = \dot{\theta} = \omega$, and $\mu = \tau_m$. With the damping torque satisfying $b(0) = 0$, the angular equation is:

$$ml^2\ddot{\theta} + b(\dot{\theta}) + mgl \sin(\theta) = \tau_m$$

Take $g = 9.81 \text{ m s}^{-2}$. At equilibrium, $\dot{\theta}^* = \ddot{\theta}^* = 0$, so:

$$\tau_m^* = mgl \sin(\theta^*) = 19.62 \sin\left(\frac{\pi}{3}\right)$$

Thus, $\tau_m^* = 16.9914 \text{ N m}$. ■

Define the output deviation $y = \theta - \frac{\pi}{3}$ and input deviation $u = \tau_m - \tau_m^*$. Keeping first-order terms gives:

$$ml^2\ddot{y} + b'(0)\dot{y} + mgl \cos(\theta^*)y = u$$

Substituting known values:

$$4\ddot{y} + 8\dot{y} + 9.81y = u \rightarrow \ddot{y} + 2\dot{y} + 2.4525y = \frac{u}{4} \quad \blacksquare$$

b) Free response about trim

Holding the input at trim means $u = 0$. Comparing the homogeneous equation with $\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2y = 0$:

$$\omega_n = \sqrt{2.4525}, \quad \zeta = \frac{1}{\sqrt{2.4525}}, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2} = \sqrt{1.4525}$$

The poles are $-1 \pm j\sqrt{1.4525}$, so the response is an exponentially decaying oscillation. For release from rest with $y(0) = \varepsilon$:

$$\theta(t) = \frac{\pi}{3} + \varepsilon e^{-t} \left(\cos(\omega_d t) + \frac{1}{\omega_d} \sin(\omega_d t) \right) \quad \blacksquare$$

Here t is in seconds. The oscillation frequency is $\omega_d \approx 1.2052 \text{ rad s}^{-1}$ (approximately 0.1918 Hz), and $\zeta \approx 0.6386$. Using the usual 2% settling-time estimate:

$$t_s \approx \frac{4}{\zeta\omega_n} = 4 \quad (\text{seconds}) \quad \blacksquare$$

The 2% band is relative to the initial displacement, and this formula is an estimate rather than the exact last band crossing.

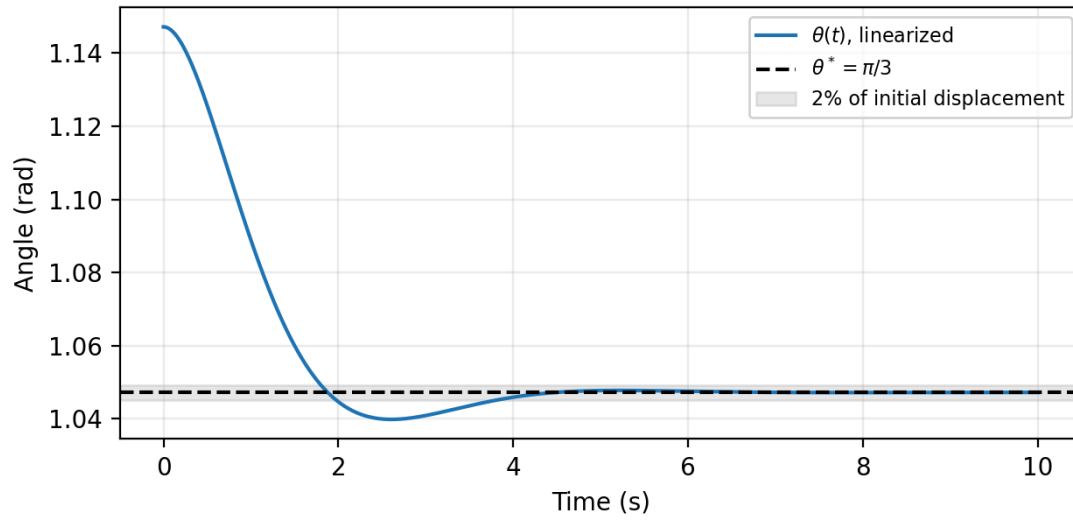


Figure 1: Linearized release response with $\varepsilon = 0.1$ rad and zero initial angular speed.

The Python calculation gives:

```

1 Trim torque = 16.991418 N m
2 Natural frequency = 1.566046 rad/s
3 Damping ratio = 0.638551
4 Damped frequency = 1.205197 rad/s
5 2% settling-time estimate = 4.000000 s

```

Problem 3

Consider a kinematic model of a unicycle maneuvering in a plane:

$$\dot{p}_x = v \cos(\theta)$$

$$\dot{p}_y = v \sin(\theta)$$

$$\dot{\theta} = \omega$$

Where (p_x, p_y) describes the location of the unicycle, and θ is its orientation; linear velocity v and angular velocity ω are regarded as the “inputs” to this kinematic model.

- Show that $\omega(t) = v(t) = 1$, $p_x(t) = \sin(t)$, $p_y(t) = 1 - \cos(t)$, and $\theta(t) = t$ is a valid solution of these equations.
- Determine the linear state equations for perturbations away from the reference solution in Part A. Is the linearization time-invariant or time-varying?
- Repeat Part B but using instead the following alternate choice of state variables:

$$z_1 = p_x \cos(\theta) + (p_y - 1) \sin(\theta)$$

$$z_2 = -p_x \sin(\theta) + (p_y - 1) \cos(\theta)$$

$$z_3 = \theta$$

HINT: You'll need to first find the nonlinear state equation corresponding to this choice of state variables, then apply the linearization process in these new coordinates.

- Verification of the reference solution

Substituting $v^* = \omega^* = 1$ and $\theta^* = t$ into the right-hand sides:

$$v^* \cos(\theta^*) = \cos(t) = \frac{d \sin(t)}{dt} = \dot{p}_x^*$$

$$v^* \sin(\theta^*) = \sin(t) = \frac{d(1 - \cos(t))}{dt} = \dot{p}_y^*$$

$$\omega^* = 1 = \frac{dt}{dt} = \dot{\theta}^*$$

All three equations hold, with initial state $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$. Thus, the stated trajectory is a valid solution. ■

- Linearization in the original coordinates

Define the state and input deviations:

$$\mathbf{x} = \begin{bmatrix} p_x - \sin(t) \\ p_y - (1 - \cos(t)) \\ \theta - t \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} v - 1 \\ \omega - 1 \end{bmatrix}$$

Taking the state and input Jacobians along the reference trajectory gives:

$$\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x} + \mathbf{B}(t)\mathbf{u}$$

where

$$\mathbf{A}(t) = \begin{bmatrix} 0 & 0 & -\sin(t) \\ 0 & 0 & \cos(t) \\ 0 & 0 & 0 \end{bmatrix}, \quad \mathbf{B}(t) = \begin{bmatrix} \cos(t) & 0 \\ \sin(t) & 0 \\ 0 & 1 \end{bmatrix}$$

The coefficients depend explicitly on time, so this linearization is time-varying. ■

c) Linearization in the alternate coordinates

Differentiate z_1 and substitute the original equations:

$$\begin{aligned} \dot{z}_1 &= \dot{p}_x \cos(\theta) + \dot{p}_y \sin(\theta) + \dot{\theta}(-p_x \sin(\theta) + (p_y - 1) \cos(\theta)) \\ &= v + \omega z_2 \end{aligned}$$

Similarly,

$$\begin{aligned} \dot{z}_2 &= -\dot{p}_x \sin(\theta) + \dot{p}_y \cos(\theta) - \dot{\theta}(p_x \cos(\theta) + (p_y - 1) \sin(\theta)) \\ &= -\omega z_1 \\ \dot{z}_3 &= \omega \end{aligned}$$

Along the reference solution:

$$\mathbf{z}^*(t) = \begin{bmatrix} 0 \\ -1 \\ t \end{bmatrix}$$

Define $\boldsymbol{\eta} = \mathbf{z} - \mathbf{z}^*$ and retain the same input deviations as above. The exact deviation equations are:

$$\begin{aligned} \dot{\eta}_1 &= \eta_2 + u_1 - u_2 + u_2 \eta_2 \\ \dot{\eta}_2 &= -\eta_1 - u_2 \eta_1 \\ \dot{\eta}_3 &= u_2 \end{aligned}$$

Dropping products of deviations:

$$\dot{\boldsymbol{\eta}} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \boldsymbol{\eta} + \begin{bmatrix} 1 & -1 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{u} \quad \blacksquare$$

Both matrices are constant, so the transformed linearization is time-invariant, even though $z_3^* = t$ varies with time.

Problem 4

A bead is constrained to move on a vertical, circular hoop of radius R . There is non-negligible friction between the bead and the hoop, creating some damping in the motion of the bead. The hoop itself is spinning about its vertical axis at a constant angular rate of $\Omega \text{ rad s}^{-1}$. Let θ be the angular position (rad) of the bead on the hoop, measured from the local vertical (down), and let $\omega = \frac{d\theta}{dt}$ be the rate of change of this angle (rad s^{-1}). The differential equation modeling the motion of the bead is then

$$\dot{\omega}(t) + b\omega(t) + a^2 \sin(\theta(t)) = \Omega^2 \cos(\theta(t)) \sin(\theta(t))$$

where $b > 0$ is a proportional damping coefficient, and $a^2 = \frac{g}{R}$ where g is the gravitational acceleration (m s^{-2}).

- Suppose that $\Omega^2 < a^2$. Show that, like the simple, unforced ($\mu^* = 0$) pendulum example considered in class, $\theta^* = 0$ and $\theta^* = \pm\pi$ are the only possible equilibrium points for this system.
- Show that the linearized dynamics of the bead angle for small displacements from the equilibria considered in Part A have an $\mathbf{A}$ matrix of the form:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}$$

Give an explicit expression for k in terms of a and Ω for each of the equilibrium points in Part A. (There is no input, hence no $\mathbf{B}$ matrix in this problem.)

- If the bead is started sliding with a small initial speed at an initial angle near, but not at, each of the equilibrium angles in Part A, will the bead tend to return to, or diverge from, that angle? Explain your answers.

a) Equilibrium points

At equilibrium, $\omega^* = 0$ and $\dot{\omega}^* = 0$. Thus:

$$\sin(\theta^*)(a^2 - \Omega^2 \cos(\theta^*)) = 0$$

When $\Omega^2 < a^2$, the second factor is strictly positive since

$$a^2 - \Omega^2 \cos(\theta^*) \geq a^2 - \Omega^2 > 0$$

Therefore, $\sin(\theta^*) = 0$. In the principal interval $[-\pi, \pi]$, the only equilibrium angles are:

$$\theta^* = 0, \pm\pi, \quad \omega^* = 0 \quad \blacksquare$$

Angles are understood modulo 2π ; π and $-\pi$ describe the same physical position.

b) Linearized dynamics

Write the nonlinear state equations:

$$\dot{\theta} = \omega, \quad \dot{\omega} = -b\omega - a^2 \sin(\theta) + \Omega^2 \cos(\theta) \sin(\theta)$$

For $\mathbf{x} = \begin{bmatrix} \theta - \theta^* \\ \omega \end{bmatrix}$, the Jacobian gives:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}$$

where

$$k = a^2 \cos(\theta^*) - \Omega^2 \cos(2\theta^*)$$

Evaluating at each equilibrium:

$$k = \begin{cases} a^2 - \Omega^2 & \theta^* = 0 \\ -a^2 - \Omega^2 & \theta^* = \pm\pi \end{cases} \quad \blacksquare$$

c) Local stability

The characteristic equation and eigenvalues are:

$$\lambda^2 + b\lambda + k = 0, \quad \lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4k}}{2}$$

At the bottom, $k = a^2 - \Omega^2 > 0$. Since $b > 0$, both eigenvalues have negative real parts. Small initial angle and speed deviations decay, and the bead returns to the bottom. The response is underdamped, critically damped, or overdamped according as b^2 is less than, equal to, or greater than $4k$. $\blacksquare$

At the top, $k = -a^2 - \Omega^2 < 0$. The eigenvalues have opposite signs, so this equilibrium is an unstable saddle. Generic nearby initial conditions diverge from the top. Only initial states on its stable manifold approach it; this exceptional set does not make the equilibrium stable. $\blacksquare$

Problem 5

Suppose in Problem 4 that the hoop is spinning faster so that $\Omega^2 > a^2$.

- Show that bead can have new equilibrium points θ^* in addition to those identified in Problem 4. Can any of these new equilibrium points satisfy $\theta^* > \frac{\pi}{2}$ (rad)? Why or why not?
- Show that the $\mathbf{A}$ matrix for the linearized dynamics about each of the equilibrium points in Part A has the same structure identified above, and determine the new value for k in terms of a , Ω , and θ^* .
- Suppose the bead has started sliding with a small initial speed and an initial angle near, but not at, zero (i.e. the bead starts near the bottom of the hoop). Will the bead tend to slide down to the bottom of the hoop (i.e. converge to the $\theta^* = 0$ equilibrium)? Compare your answer with Problem 4, Part C.
- As a function of a , find the value of Ω such that the bead will have $\theta^* = \frac{\pi}{3}$ as an equilibrium point.
- Suppose the bead has started sliding with a small initial speed and an initial angle near $\frac{\pi}{3}$. Will the bead tend to converge back to a steady-state angle of $\frac{\pi}{3}$? Explain your answer.

a) Additional equilibria

The equilibrium condition remains:

$$\sin(\theta^*)(a^2 - \Omega^2 \cos(\theta^*)) = 0$$

In addition to 0 and $\pm\pi$, the second factor now admits:

$$\cos(\theta^*) = \frac{a^2}{\Omega^2}, \quad \theta^* = \pm \arccos\left(\frac{a^2}{\Omega^2}\right) \quad \blacksquare$$

Since $0 < \frac{a^2}{\Omega^2} < 1$, the new angles satisfy $0 < |\theta^*| < \frac{\pi}{2}$ in the principal interval. Thus, neither new physical equilibrium lies above the horizontal. Adding multiples of 2π can produce numerical angles greater than $\frac{\pi}{2}$, but these describe the same positions.

b) Linearized dynamics

Using the Jacobian found in Problem 4:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -k & -b \end{bmatrix}, \quad k = a^2 \cos(\theta^*) - \Omega^2 \cos(2\theta^*)$$

At either new equilibrium, substitute $a^2 = \Omega^2 \cos(\theta^*)$:

$$k = \Omega^2 \sin^2(\theta^*) = \Omega^2 - \frac{a^4}{\Omega^2} > 0 \quad \blacksquare$$

For completeness, the original equilibria still have $k(0) = a^2 - \Omega^2 < 0$ and $k(\pm\pi) = -a^2 - \Omega^2 < 0$.

c) Motion near the bottom

At $\theta^* = 0$, $k < 0$, so the linearization has one positive and one negative eigenvalue. The bottom is now an unstable saddle: generic small disturbances grow, and the bead does not return to the bottom. As with the top in Problem 4, approach along the stable manifold is exceptional. This differs from Problem 4, where $\Omega^2 < a^2$ made the bottom locally asymptotically stable. $\blacksquare$

d) Rotation rate for the specified equilibrium

Set $\theta^* = \frac{\pi}{3}$:

$$a^2 = \Omega^2 \cos\left(\frac{\pi}{3}\right) = \frac{\Omega^2}{2} \rightarrow |\Omega| = \sqrt{2}a$$

Taking the positive rotation rate gives $\Omega = \sqrt{2}a$ ■. The opposite rotation direction gives the same equilibrium.

e) Motion near $\frac{\pi}{3}$

At the rate found above:

$$k = \Omega^2 \sin^2\left(\frac{\pi}{3}\right) = 2a^2 \cdot \frac{3}{4} = 3\frac{a^2}{2} > 0$$

With $b > 0$, both eigenvalues have negative real parts. Thus, sufficiently small initial angle and speed deviations converge back to $\frac{\pi}{3}$. This is a local conclusion about the nonlinear system, obtained from its asymptotically stable linearization. ■

Problem 6

The equations of motion given in Problem 4 are valid even if the hoop rotation rate Ω is changing with time. Suppose that the nominal rotation rate Ω^* of the hoop is sufficient that the system $\theta^* = \frac{\pi}{3}$ as an equilibrium point, but now we will directly vary the hoop speed $\Omega(t)$ slightly around this nominal value to influence the motion of the bead.

- Treating the hoop spin rate $\Omega(t)$ as the input to the system, determine the linearized dynamics of the motion of the bead for small displacements from the $\theta^* = \frac{\pi}{3}$ equilibrium. (You will now have both $\mathbf{A}$ and $\mathbf{B}$ matrices.)
- Suppose the bead has started sliding with a small initial speed and an initial angle near $\frac{\pi}{3}$. We would like to dynamically vary the speed of the hoop so that the bead will settle back to the $\theta^* = \frac{\pi}{3}$ equilibrium with transient oscillations of 2 rad s^{-1} and a damping ratio of 0.7. Assuming the bead angle and angular rate of change are measured, specify a feedback control law for $\Omega(t)$ which will achieve these objectives.

HINT: A simple PD control strategy for the linearization should be sufficient; for simplicity, you may assume the numerical values $a = 3$ and $b = 0.25$. *Be careful to distinguish $\mu(t) = \Omega(t)$ from $u(t)$ in your analysis!*

- Linearization with hoop speed as input

Choose the positive nominal rate $\mu^* = \Omega^* = \sqrt{2}a$. Define:

$$\mathbf{x} = \begin{bmatrix} \theta - \frac{\pi}{3} \\ \omega \end{bmatrix}, \quad \mu(t) = \Omega(t), \quad u(t) = \mu(t) - \mu^*$$

The nonlinear angular acceleration is:

$$f(\theta, \omega, \mu) = -b\omega - a^2 \sin(\theta) + \mu^2 \cos(\theta) \sin(\theta)$$

The input derivative at trim is:

$$\left. \frac{\partial f}{\partial \mu} \right|_* = 2\Omega^* \cos\left(\frac{\pi}{3}\right) \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2} \Omega^* = \frac{\sqrt{6}}{2} a$$

Using $k = 3\frac{a^2}{2}$ from Problem 5:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u, \quad \mathbf{A} = \begin{bmatrix} 0 & 1 \\ -3\frac{a^2}{2} & -b \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 \\ \sqrt{6}\frac{a}{2} \end{bmatrix} \quad \blacksquare$$

This is an equation in the input deviation u , not the full hoop rate μ .

- PD feedback design

Interpret the specified transient oscillation frequency as the damped frequency $\omega_d = 2 \text{ rad s}^{-1}$. For $\zeta = 0.7$:

$$\omega_n = \frac{\omega_d}{\sqrt{1 - \zeta^2}} = \frac{2}{\sqrt{0.51}}, \quad \omega_n^2 = \frac{400}{51}$$

The desired characteristic polynomial is:

$$\lambda^2 + 2\zeta\omega_n\lambda + \omega_n^2 \approx \lambda^2 + 3.920784\lambda + 7.843137$$

Let $\beta = \sqrt{6}\frac{a}{2}$ and choose:

$$u = -K_p x_1 - K_d x_2$$

Then

$$\ddot{x}_1 + (b + \beta K_d) \dot{x}_1 + \left(3 \frac{a^2}{2} + \beta K_p\right) x_1 = 0$$

Matching coefficients gives:

$$K_p = \frac{\omega_n^2 - 3 \frac{a^2}{2}}{\beta}, \quad K_d = \frac{2\zeta\omega_n - b}{\beta}$$

With $a = 3$ and $b = 0.25$, the Python calculation gives:

```
1 Nominal hoop rate = 4.242641 rad/s
2 Input coefficient = 3.674235
3 Desired natural frequency = 2.800560 rad/s
4 Kp = -1.539603 s^-1
5 Kd = 0.999061
6 Closed-loop poles = -1.960392+2.000000j, -1.960392-2.000000j s^-1
```

Thus, the commanded physical hoop speed is:

$$\Omega(t) = \Omega^* - K_p \theta(t) - \frac{\pi}{3} - K_d \omega(t)$$

Numerically, with angles in radians and time in seconds:

$$\Omega(t) \approx 4.242641 + 1.539603 \left(\theta(t) - \frac{\pi}{3} \right) - 0.999061 \omega(t) \quad \blacksquare$$

The proportional gain K_p is negative because the requested stiffness ω_n^2 is smaller than the uncontrolled stiffness $3 \frac{a^2}{2} = 13.5$. The derivative term increases damping. The resulting poles are $-1.960392 \pm 2j$, which give the specified damped frequency and damping ratio.

These targets apply to the local linearization. As a numerical check, apply the same feedback to the original nonlinear equations with $\theta(0) - \frac{\pi}{3} = 0.02$ rad and $\omega(0) = 0.01$ rad s⁻¹:

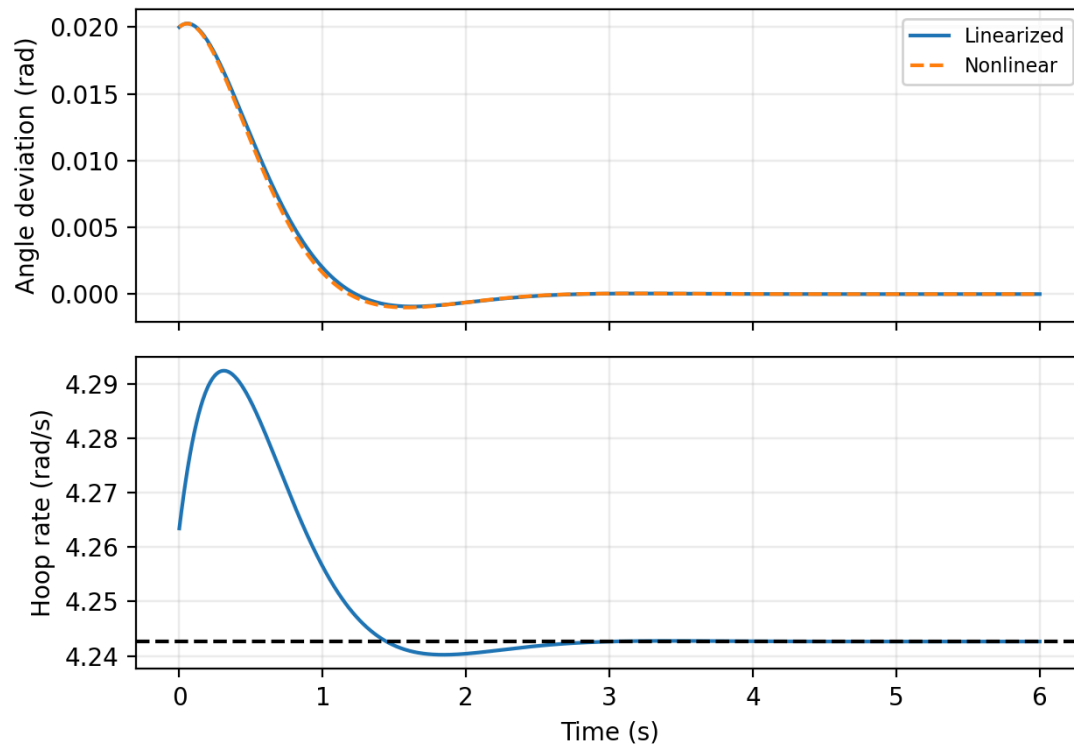


Figure 2: Linear and nonlinear responses under the PD law, and the commanded nonlinear hoop rate.

Code

```

./src/index.py Python
1  from pathlib import Path
2
3  import matplotlib
4  import matplotlib.pyplot as plt
5  import numpy as np
6  from scipy.integrate import solve_ivp
7
8  matplotlib.use("Agg")
9
10 ROOT = Path(__file__).resolve().parents[1]
11 FIGURES = ROOT / "outputs/figures"
12 TEXT = ROOT / "outputs/text"
13
14
15 def heading(problem):
16     print(f"\n-----\n {problem}\n-----")
17
18
19 def report(name, content):
20     print(content)
21     (TEXT / f"{name}.txt").write_text(content + "\n", encoding="utf-8")
22
23
24 def save(fig, name):
25     fig.tight_layout()
26     fig.savefig(FIGURES / f"{name}.png", dpi=200)
27     plt.close(fig)
28
29
30 def main():
31     FIGURES.mkdir(parents=True, exist_ok=True)
32     TEXT.mkdir(parents=True, exist_ok=True)
33
34     #####
35     # p01 #
36     #####
37     heading("p01")
38     V = np.array([[2, -2, 3], [-1, 2, 1], [4, 1, 3]])
39     assert np.allclose(V @ (np.array([29, 53, -15]) / 31), [-3, 2, 4])
40     print("Verified the supplied change-of-basis solution.")
41
42     #####
43     # p02 #
44     #####

```

```

45     heading("p02")
46     g, length, mass, damping = 9.81, 2.0, 1.0, 8.0
47     trim = np.pi / 3
48     inertia = mass * length**2
49     torque = mass * g * length * np.sin(trim)
50     wn = np.sqrt(g / length * np.cos(trim))
51     sigma = damping / (2 * inertia)
52     zeta = sigma / wn
53     wd = np.sqrt(wn**2 - sigma**2)
54     t = np.linspace(0, 10, 2001)
55     offset = 0.1 # illustrative small release displacement, rad
56     deviation = (
57         offset * np.exp(-sigma * t) * (np.cos(wd * t) + sigma / wd * np.sin(wd * t))
58     )
59     report(
60         "s02",
61         f"Trim torque = {torque:.6f} N m\n",
62         f"Natural frequency = {wn:.6f} rad/s\n",
63         f"Damping ratio = {zeta:.6f}\n",
64         f"Damped frequency = {wd:.6f} rad/s\n",
65         f"2% settling-time estimate = {4 / sigma:.6f} s",
66     )
67     fig, ax = plt.subplots(figsize=(6.4, 3.2))
68     ax.plot(t, trim + deviation, label=r"$\theta(t)$, linearized")
69     ax.axhline(trim, color="black", linestyle="--", label=r"$\theta^*=\pi/3$")
70     ax.axhspan(
71         trim - 0.02 * offset,
72         trim + 0.02 * offset,
73         alpha=0.18,
74         color="gray",
75         label="2% of initial displacement",
76     )
77     ax.set(xlabel="Time (s)", ylabel="Angle (rad)")
78     ax.grid(alpha=0.25)
79     ax.legend(fontsize=8)
80     save(fig, "s02")
81
82     #####
83     # p03 #
84     #####
85     heading("p03")
86     print("Reference in transformed coordinates: z* = [0, -1, t].")
87
88     #####
89     # p04 #
90     #####

```

```

91     heading("p04")
92     print("Omega^2 < a^2: bottom asymptotically stable; top a saddle.")
93
94     #####
95     # p05 #
96     #####
97     heading("p05")
98     print("Omega^2 > a^2: new equilibria +/- arccos(a^2/Omega^2) stable.")
99
100    #####
101    # p06 #
102    #####
103    heading("p06")
104    a, b = 3.0, 0.25
105    nominal = np.sqrt(2) * a
106    k = 1.5 * a**2
107    gain = np.sqrt(3) * nominal / 2
108    desired_wd, desired_zeta = 2.0, 0.7
109    desired_wn = desired_wd / np.sqrt(1 - desired_zeta**2)
110    kp = (desired_wn**2 - k) / gain
111    kd = (2 * desired_zeta * desired_wn - b) / gain
112    Acl = np.array([[0.0, 1.0], [-k - gain * kp, -b - gain * kd]])
113    poles = np.linalg.eigvals(Acl)
114    assert np.allclose(np.abs(poles.imag), desired_wd)
115    assert np.allclose(-poles.real / np.abs(poles), desired_zeta)
116    report(
117        "s06b",
118        f"Nominal hoop rate = {nominal:.6f} rad/s\n",
119        f"Input coefficient = {gain:.6f}\n",
120        f"Desired natural frequency = {desired_wn:.6f} rad/s\n",
121        f"Kp = {kp:.6f} s^-1\nKd = {kd:.6f}\n",
122        f"Closed-loop poles = {poles[0]:.6f}, {poles[1]:.6f} s^-1",
123    )
124    t = np.linspace(0, 6, 1201)
125    initial = [0.02, 0.01]
126
127    def nonlinear(_, state):
128        error, rate = state
129        angle = trim + error
130        hoop_rate = nominal - kp * error - kd * rate
131        return [
132            rate,
133            -b * rate
134            - a * a * np.sin(angle)
135            + hoop_rate**2 * np.cos(angle) * np.sin(angle),
136        ]

```

```
137
138     linear = solve_ivp(
139         lambda _, x: Acl @ x, [0, t[-1]], initial, t_eval=t, rtol=1e-10, atol=1e-12
140     )
141     nonlinear_result = solve_ivp(
142         nonlinear, [0, t[-1]], initial, t_eval=t, rtol=1e-10, atol=1e-12
143     )
144     assert linear.success and nonlinear_result.success
145     fig, axes = plt.subplots(2, 1, figsize=(6.4, 4.5), sharex=True)
146     axes[0].plot(t, linear.y[0], label="Linearized")
147     axes[0].plot(t, nonlinear_result.y[0], "--", label="Nonlinear")
148     axes[0].set ylabel="Angle deviation (rad)"
149     axes[0].legend(fontsize=8)
150     axes[1].plot(t, nominal - kp * nonlinear_result.y[0] - kd * nonlinear_result.y[1])
151     axes[1].axhline(nominal, color="black", linestyle="--")
152     axes[1].set xlabel="Time (s)", ylabel="Hoop rate (rad/s)"
153     for ax in axes:
154         ax.grid(alpha=0.25)
155     save(fig, "s06b")
156
157
158 if __name__ == "__main__":
159     main()
```